

ActInf Livestream #006.1: "A tale of two densities: active inference is enacted"

Livestream | 54:30

hello everyone welcome to active inference live stream number 6.1 it is October 13th 2020 whether you're a firsttime listener or not welcome to tecom tecom is an experiment in online team communication and learning related to active inference you can find us on Twitter at inference active at our Gmail address our keyb team that's public or our YouTube channel this is a recorded and an archived live stream so please provide us feedback so that we can improve our work also all backgrounds and perspectives are welcome here and in service of that remember your video etiquette M mute if there's sound in the background and raise your hand so that we'll be able to hear from everyone here we are in actim stream 6.1 and today's stream is going to go like this we're going to start out with a warm-up section where we introduce some of ourselves especially our newer participants and just have a quick check-in then we'll turn to the discussion of the paper today we'll try to cover the goal of the paper the road map how they get from A to Z go through the abstract which is how the authors represent their work in the most distilled form and then look at some of the figures see where they're going to be going with the figures what they're going to be showing and then next week in 6.2 we're going to have a lot more

time for further discussion on this
paper and also to dive into some of the
technical details of the figures so
please save and submit your
questions all right here we are in the
intros and check-ins for the
introduction section please introduce
yourself and your location just say a
quick hello anything else you'd like to
add and then feel free to pass it to
someone else so I'm Daniel I'm in Davis
California and I'll pass it to one of
our firsttime participants
Maas it's actually matthys Matthis pink
um I'm a master student in
Germany in um at the University of osnar
Brook which is in lower Saxony so it's
uh afternoon where I am I guess it's in
the morning where you are um
yeah I'm excited about
participating thanks for having
me great let's I
will I don't know is there another
person who's new here uh lee Alex
maybe let's go to Lee our other new
participant yeah hi um yeah I'm Lee uh
I'm based in London but I'm actually
studying at the uh University of York uh
studying embod cognition and um yeah
look really looking forward to this um I
will pass on to
Alejandra hello everyone
um I'm again at Mexico kind of tired
um last week was
uh very hard but um yeah nice to be here
again with you all and um pass it
to um
Alex uh hello everybody I'm Alex

I'm an engineer basically and now also a
researcher and I am situated in Moscow
Russia and affiliated with uh system
management
school so I'm pass it to
clip hi I'm John clip I'm in the
Cambridge Mass with the MIT media lab
I'm actually doing this at in Northern
New Hampshire at my farm um and I'm very
pleased to be a part of this and here to
[Music]

learn let's go to Alex kefir I'll pass
it off to
Maxwell

okay hi everyone uh so I was here a
couple weeks ago um I'm a philosopher I
guess and sometime computational modeler
at manash University and right now I'm
in New York City um and I've been
involved in this debate uh that's that's
explored in this paper for a while so
thought I would join in and I will pass
it to who's left um

Shannon hi I'm Shannon Crooks um I'm
part of the sensory motor Neuroscience
Lab at the University of California in
mered but currently I'm in South Dakota
um so I'll pass it
to

Sasha hi everybody I'm Sasha I'm based
out of Davis California and I'm
affiliated with University of California
Davis and also um just very excited to
learn and unpack some of these topics um
it's been a really great uh past six or
so streams
so I will pass it on to Stephen
[Music]

uh hello yeah I've been involved in this
for the last few sessions as well find
it very very helpful and very
interesting I'm um doing a
practice-based PhD into some um
processes to explore social
topographies um and uh at Christ
Church University at the Solomon School
of Applied
psychology um with the help of the
professional development
Institute um and I'll pass that to
Maxwell I suppose is it I believe I'm
the last one um yeah so uh I'm Maxwell
ramstead uh I'm based in Montreal where
I'm talking to you from uh at Mill
University and I'm also the first author
on the paper that we'll be discussing
today so I'm quite excited to discuss it
with you uh thanks Alex kefir by the way
for showing up Alex and I have had a
sustained
series of discussions around these
papers that have led to followup papers
and I think to a really robust and fun
friendship uh through these discussions
so I'm looking forward to discussing all
these issues with you uh thanks uh
thanks for the uh continued following of
our
work awesome you know we bring the ideas
together sometimes through the people
and the friendships so and I guess I'm
not going to pass it to anyone since
since everyone else has uh spoken right
yep I think that is everyone so cool
let's go to our warm-up questions and
anyone who likes to speak can uh feel

free to raise their hand or jump in if there's no one Speaking yet so the first question is what Drew you to this paper or topic and while people are raising their hand I'll start I think what was exciting about this paper was about just combining two different schools of thought and bringing them together in a constructive way that was really super additive rather than just choosing sides what about you

Maxwell um well so the motivation for writing the paper in part um was that I don't think that at least at the time I didn't think that what the generative model business uh you know under the free energy principle is all about really I don't think that was well understood um at the time um that we wrote this so I I I allude to this I think a few weeks ago uh this paper uh the answering Schrodinger's question paper in physics of Life reviews and the paper that we discussed the last few weeks multiscale integration were all originally the same paper it was all one big thing and we ended up splitting it into uh different papers to address different things uh so you know the multiscale stuff that we'd been uh discussing to address like the the the scale-free and multi-level formulation of the Fe and the the aim here was really to connect uh the Fe to other pragmatist approaches uh the center act the center on action um and also to clarify uh the nature of the generative

models that are at play here uh which
aren't just kind of brain bound
statistical models but turn out to be
something like the phenotype of the
organism cool anyone else have thoughts
on
that uh Alex
kefir yeah thanks um so I guess well I
was drawn into the debate because um um
some of my work with yakob hovi was
among the uh the sort of Target um the
critical Target of the article although
Maxwell and Company were very nice to us
and they didn't say we were wrong they
just said this doesn't exactly
generalize to the free energy principle
um so that's how I got drawn into this
and I continue to be interested in it
because I think it I mean I think we've
we've converged on more of an agreement
than maybe existed at this time this
paper came out but oh yeah 100% I think
I think we we've basically converged to
one coherent story at this point but
this is two years in the making so yeah
but but I mean the issues raised here
are still interesting to think about you
know and I continue to learn more by
thinking about them so awesome all right
the second warm-up question is going to
be what would be something that you'd
like to have resolved by the end of
today's discussion so that could be a
specific question about how to apply
something to a system or Le would like
to speak yeah Lee go
ahead go ahead Lee H sorry I was just
turning my sound up um so so yeah

essentially um I I guess I I was drawn to this topic because um uh my journey to towards active inference has been via cognitive Linguistics so I started off looking into relationships between language and perception and particularly metaphor so I think I've got quite a different understanding of um what's meant by model uh probably something more kind of phenomenological and uh maybe epistemic um and and I really starting to understand that that's not what really what is meant by active inference so I've taken a good look through this paper and I'm hoping to build some kind of bridge from uh where I'm at to more of an understanding of what's implied by generative model in active influence cool well great because that's one of the main topics of the paper and Stephen actually just so just bouncing off what was just said there in some ways when you take metaphors if you bring metaphors into embodiment and as being embodied in space and around us actually then suddenly that whole metaphor work becomes actually almost very close to what might be a generative model in a funny way even though it's different so there might be some way I'm actually so in some ways I'm kind of interested how this sort of becomes full circle and can come back to these kind of um uh kind of human ways of knowing even though we've accessing it through um maybe kind of of abstracted

models interesting and I think we'll return to some of these ideas if anyone else wants to chime in on what they'd like to have resolved but very interesting to hear about metaphor as cognition and how we can humanize our understanding of some of these technical or conceptual issues all right well if there are no hands remaining I'll move on to the next slide So today we're going to going to be talking about A Tale of Two densities active inference is an active inference which is a article in adaptive behavior in 2020 by Maxwell ramstead kiroff and friston and in this paper they lay out their goal really clearly which I always love to see in a paper they write at the very beginning the aim of this article is to clarify how best to interpret some of the central constructs that underwrite the free energy principle or FP and its Cory active inference in theoretical neuroscience and biology namely the role that generative models and recognition densities play in this Theory aiming to unify life and mind so what are the two densities they're going to be the generative models and the recognition densities and we're going to learn more about them and hear about how they're related and discuss different perspectives on how the densities are linked and specifically the question is what is the tale of these two densities it's alluded to in in the title and it's awesome that we have Maxwell and Alex and so many other voices here to to make

that synthesis and that tale that we're all telling together realized so um in the ACT imp stream 6.0 I provided a little bit of context just from my perspective on some of these issues if people want to learn more about uh some of the background ideas but for now we're going to just jump into the abstract and at any point people can just raise their hand and I'll just pause right there and we'll take a comment or a thought in the abstract they begin by rehearsing what I had just read that they're looking to clarify how to best interpret some of the constructs underwriting the free energy principle and those two constructs are the generative models and the variational densities so those are the two densities and the tail is going to link them we argue that these constructs generative models and variational densities have been systematically misrepresented in the literature because of the conflation between the Fe and active inference on one hand and distinct albeit closely related basian formulations centered on the brain variously known as predictive processing predictive coding or the prediction error minimization framework more specifically we examine two contrasting interpretations of these active inference type models a structural representationalist interpretation and an inactive interpretation so so we're setting up the two sort of sub stories these are

the the tension between these two perspectives that we're going to be looking to resolve under the F through active inference we argue that the structural representationalist interpretation of generative and recognition models does not do justice to the role that these constructs play in active inference under the FP we propose an inactive interpretation of active inference what might be called an active inference in active inference under the FP the generative and recognition models are best cast as realizing inference and control the self-organizing belief guided selection of action policies and do not have the properties ascribed by structural representationalists so for this next slide I'll really appreciate anyone's um perspective or linking it back to things they've seen before because I just put it up there as a visual and uh just a starting place on the left side here we have the Bas structural representationalist perspective that we're just trying to highlight the features that are going to be most simple to carry forward and the basian structural representationalist story is about how data and another type of data which are often called hyperparameters are linked through a recognition model that takes data like sensory data and recognizes it and then um going the other direction you have the hyperparameters that are generating sensory data and we can also to talk

about why it's important to have this generative step and the outcome of this um basian computationalist scheme is that there's a statistical conversion of a multi-level model that represents structures of the world through something like expectation maximization or em models and we can contrast that with the inactive Paradigm and the inactive Paradigm is about how agents and the world are related through perception and action and the outcome of the inactivist perspective is in embodied ecological action sequence really from an embedded agent who is enacting behavior and so this is always the school of thought where we see all the ease encultured and embedded and all these things and they um seemingly at least at the first pass up to two quite different explanatory uh outcomes they seem to be talking about somewhat different aspects about the world and they definitely link them through different ways so I'm just curious um Maxwell or anyone else like what led to these two models being the two cities the two densities that were chosen how does one come down to just two why is there not one or three cities and then how did the basian structural representationalist and the inactive Viewpoint rise up as like the two kind of uh tier one theories that we wanted to find a synthesis between well so when I got into this literature um especially from the the vantage point of philosophy what I

noticed was that very very little of it was technically rigorous in the sense that you know a lot of it was telling uh like a story about how you know the brain roughly performs bayes and inference and then uh you know kind of saying well there's like a family of different theories that do this in various ways and grouping the free energy principle under that I say A you know uh there was like a lack of technical rigor uh I I want to emphasize that Alex's papers uh with yakob are probably the exception to that uh when I uh consulted Alex's papers in 2018 I thought well here's you know some wonderful work that is really taking the time to drill down on um the formalisms as they're used to study the brain I thought that was great uh but i' I spent a lot of time really drilling into the formalism of the free energy principle per se and one of the things that I was sort of surprised to find out as I was learning the formalism was that although everyone is talking about the generative model uh there are really two models at play um so those are the generative and the recognition models or densities equivalently um I mean so first of all we say model by model we just mean a probability density right so uh a probability distribution over a bunch of variables uh that are of interest to us um and so yeah the the two models in question uh under the free energy principle function slightly differently than they do in more um traditional

brain-based uh so to step back a bit in
in um in machine learning and in
statistics a recognition model basically
tells you the probability of some State
given a bunch of other things it's a
it's not a joint probability
distribution um it it and it's it's used
essentially to recognize what's causing
your data so you're using it basically
to invert uh your mapping I mean Alex
also if I'm saying anything uh
inaccurate here just uh please uh jump
in and let me know uh but yeah so
there's basically in traditional uh kind
of you know basic and brain machine
learning architectures the generative
and the recognition models are basically
just the inverse of one another so uh
the kind of top uh the kind of bottom up
pass uh is a recognition uh pass it's a
recognition model in the sense that it's
starting from the data and then kind of
passing through uh the network you're
able to infer what must have caused your
data and uh the generative model is the
inverse path where you're starting from
your beliefs about States and you can
generate fictive data you know based on
uh on this
model and at least in AI papers the way
that this uh well the this is how it was
described as applying to uh the free
energy formulation as well and my point
in this paper was that well this is a a
very technically rigorous and accurate
description of what's going on in the
basian brain but these constructs have a
slightly different meaning under active

inference uh and to bring it back to
I'm almost done uh just give me one one
more minute uh to bring it back to uh
these schemas basically the so the
recognition density is sort of like your
best guess right now you can think about
it sort of like as your posterior and
your prior so your recognition density
is a density defined over all of your
States and your parameters and it
basically tells you what do I believe is
the most probable value of these states
and parameters now you know given uh my
prior beliefs in my evidence your
generative model uh to the contrary is
the point of reference
uh you know for the generation of free
energy gradients so it's not your
posterior it's your prior it's sort of
like it harnesses all of the priors um
especially the priors about your
preferred data
distributions relative to which the free
energy and therefore the Dynamics are
defined so you know you write uh
inference and control I think you
discussed uh in 6.0 Dan yeah the so the
recognition model is responsible for
inference and the generative model is
responsible for control you might say so
I'll stop there cool Alex keeper and
then we'll go to anyone else with a
raised hand yes thanks so I I stopped
myself from jumping in I mean that was
that was a good summary um the only
thing the point at which I wanted to
jump in was to say well so the
recognition model isn't uh as Con

constru in these Basi and brain theories
is is an approximate inversion of the
generative model so my my um if I have
any complaint about this paper it's that
I think that there's a a closer sort of
conceptual connection between the
generative and recognition densities
than than maybe the paper suggests um
and that I don't think you can cleanly
separate these things so uh well anyway
I don't want to launch into this yet
what I wanted to do first was just
address the the question was sort of how
did these two visions sort of arise and
my sense is that what Carl friston did
he did a lot of great stuff but the main
thing he did that distinguish his
approach from the existing stuff in
machine learning a lot of which was
based on free energy minimization uh
which I think he came to around the same
time as people like Jeff Hinton but
anyway he you know he added action into
the picture and he pointed out that you
can act so as to reduce the surprisal or
the free energy caused by your sensory
States uh instead of just revising your
generative generative distribution um
anyway so I think I'll hold off on on
arguing for the moment awesome just want
to say I agree with you now uh I the
these two things do do not really come
together uh come apart sorry um I mean
it's it's really just an implementation
of variational inference from that point
of view and variational inference and
this is why it's a tale of two densities
is that to do this variational inference

thing you need both you need a kind of point of reference that's going to give you the free energy gradients and then you need a sort of what is my best guess as I am performing gradient descent on my free energy

and yeah like I I I don't want I don't want to say that the two come apart and indeed I mean I think you know in in a newer paper that I've written on this uh I basically just straight up say that you were correct initially uh in with respect to the recognition density so the recognition density it's fair to say that it's a representation in the structural representationalist sense that uh you've been uh articulating in in in the series of really awesome papers which you should all read by the way I think

uh yeah whereas the the the one point of disagreement that I think we've clarified now is the status of these generative models and uh yeah yeah I'll I'll stop there too nice we'll go to Stephen and then anyone else with a raised hand before the road map yeah one question I've got is in terms of if if these um kind of um hyper parameters that they use in machine learning are are using a free energy is it that they're they're just minimizing the kind of energy expenditure and the entropy internally and and they're not using Shannon entropy I they're not looking at how entropy is inferred or transmitted sort of in a second order

process from interacting in the world
they're just minimizing it within the
kind of data that's being kind of ACR
which kind of in some ways seems to
happen easily when you look at like
Vision data but may not be so easy to
pass when you sort of look at the whole
body but that that question of entropy
is it that they use entropy in a
different way and they're not really
using entropy in the kind of um external
entropy of interaction but kind of just
the
entropy within the um calculations
in terms of there are two kinds of
entropy at play in general right uh so
entropy in the thermodynamic sense is a
measure of how many macro states are com
uh compatible with a given uh value of a
macro state right so I don't know your
temperature is 36 degrees Celsius how
many different macro State
configurations are compatible with
that um so that form of entropy is one
special case of the broader kind of
entropy which is more or less a measure
of how flat your probability
distribution is uh over your States so
if you have a perfectly flat
distribution your entropy is optimal and
um so the the entropy that we're
concerned with is really the second type
it's the information theoretic entropy
it but uh it's it's transpired over the
last few years that uh variational free
energy is also a thermodynamic free
energy you just have to mult mly by
boltzman's constant so essentially yeah

it's always information theoretic measures but when this is realized in actual physical systems it is also a thermodynamic uh free energy but you know the the variational free energy it sounds kind of spooky and esoteric but it's really simple really is that you have a preferred data distribution and you have an actual data distribution and the free energy is just a way to quantify the difference between the two um yeah awesome we'll go to to Alex and then back to

Stephen yeah I was just going to briefly say in the you know the earlier machine learning literature the free energy definitely wasn't supposed to be anything thermodynamic or I mean maybe that was it maybe that was an open possibility but it was just a it was just a measure of as you were saying I think Stephen between two internal internally determined distributions it was the the top- down generative posterior versus the approximate posterior and um any connections to thermodynamics are really cool but I think that's a additional like very substantive um question slth thesis Stephen and then anyone else who raised their

hand yeah I think this is this is quite a useful distinction because I think that's where a lot of the mixing gets sort of caught up at the moment because a lot of stuff has been referred to in like the last 15 years around basian optimization and basan stuff and then

where it's all pretty
much as if it's contained in the data
in the brain and then
now it's like try it's like a
reconceptualizing of that so I think it
is a big big challenge for people to
like forget about what they've learned
before really agreed with that Stephen
and um that's really what this
conversation and synthesis is about is
about bringing those qualitative off
insights from an activism and saying
well wait a minute perception isn't
simply the reverse of action you know
there's not photons coming out of your
eye so what is different between
perception and action but also
recognizing that the world and the agent
have this sort of symmetry and that they
are linked so again anyone can raise
their hand but we're going to turn to
the road map and ask how the authors
said up the discussion between the
basian and the inactivist approached and
then ultimately converged towards some
of a synthesis and on the right side is
just the title page of the Dickens work
Tale of Two Cities so first there's an
introduction sequence and then they talk
about statistical models as
representations and specifically they
talk about generative models and
recognition models in basian cognitive
science as well as generative models as
structural representations so that's the
basian structural and the
representational
components then they talk about

generative models and action policies which is sort of opening the door to this potential inadequacy of the purely basian approach which is that it's not action oriented in the third section they discuss the active inference framework first they start by talking about how phenotypes are conceptualized under active inference and about markov blankets and there's a figure one which we've seen before and we will see again about how markov blankets and active inference are linked they then discuss surprise entropy and variational free energy which are all terms uh perhaps not surprise but the other two that we've brought up in this discussion and in the paper they really go into detail a bit more about how these topics are linked there's then a discussion about how active inference specifically links the variational free energy with the inferential models in section four A Tale of Two densities they talk about how the generative model and the recognition density are to be conceptualized under the Fe and specifically they talk about the generative model and the generative process in active inference in figure two they depict a generative model and represent it as a basian network in figure three they talk about the same generative model in active inference represented as a forny factor graph and the similarities and differences between figure two and three we probably going to go into next week in

6.2 there's then a section on
variational inference and recognition
Dynamics under the F in the fifth
section they turn to purely inactive
inference and they say first the claim
that the generative models are control
systems and so this sort of borrows on
and builds on the generative model idea
which is a bit of a computationalist
perspective as well as this control
systems perspective which implicitly
means action orientation because control
of action not just control of
theoretical
parameters they then take the fighting
words section generative models are not
structural representations I always like
it when we can get down to specific
claims and negatable state statements
and uh just really being specific
because sometimes these abstractions get
pretty far out there's then the final
figure four which depicts the action
perception cycle in active inference as
a generative model and a process and
last they conclude with some remarks and
the remarks also conclude with towards
multi-disciplinary research
characteristics for cognitive science
and I know we have a lot of awesome
cognitive scientists and other
perspectives in this room and so this is
sort of our springboard where we can
take the ideas that we're talking about
here and ask well how does this impact
how we're going to do the cognitive
science research what kind of
explanations what kind of predictions

what kind of um experiments are we going to do differently how will our research Paradigm be different if we take this approach that is synthesizing the inactive and the basian approach so especially cognitive scientists um would be awesome to hear any thoughts and while people are sort of collecting their thoughts about anything on the road map or prior in the uh last 20 minutes here we'll have good time to go through the figures and ask about how they relate to this big topic how are the two cities linked what is the highway that connects the two cities or what is the broader collaborative Network that links these two any thoughts here on the road map you you definitely laid out the reasoning in the paper very well so thank you good and uh these papers are uh straightforward to lay out because they really have almost a uh additive way of laying out the topics first you lay out the strongest version of the other approach and actually it's the um professionalism and the clarity and not misrepresenting another Viewpoint just to throw it under the bus or to make a straw man argument it's it's laying out the strongest form of the um prior literature is what allows us to build another level of strength on top of the literature rather than just cutting out the pillars from underneath us so I always well and I have to say you know we started off with this a much stronger position

uh than we ended up with and it's it's
by engaging seriously with the arguments
that Alex and yakob had prepared uh you
know with a few other people pow glazi
and so on had been uh arguing for this
kind of stuff but I mean thank you for
for saying that about the
professionalism I I really think that
there there is something to that
argument and the whole point is to keep
the baby you know uh with while getting
rid of the bath water uh and you know
the the baby is this sort of idea that
there is something like a a rep
representations of the external world
that you know carry semantic content and
so on like uh it's just that it's not
the generative models uh it's it's the
recognition densities that's that that
was sort of The Tweak that we wanted to
bring to the table but cool Alex and
then anyone else who raises their
hand yeah so um yeah like like Maxwell
says we've had a lot of interesting
discs about this um I guess the point
the point on which I remain not
convinced yet is that I think I still
think of the generative model even in
the FP as a representation so we can we
can maybe talk about that um the I think
to me what this brings to the table and
this is like transformed how I see this
stuff is the importance of understanding
the generative model as a control system
um I just think you can do that in a way
that also grants that it's a structural
representation and that's but I've been
that's still the framework in which I

work um so I think there's another question raised here about whether the generative model is sort of encoded and um I do understand where where the authors are coming from um on that question and I think it's not so straightforward at in the FP as it would be in some earlier models like the H holds machine um so I don't know I'm curious to see if Maxwell and I still disagree I have the sense that I maybe still want to call the generative model a representation

um more than Maxell does but we can get to that can I ask um Alex and again anyone can raise their hand and have a here too is what is the advantage of wanting something to be a representation or what is the alternative to something being structural right so I mean so I don't think in those terms in terms of the advantage of it I just think is it a representation or not I think it is based on what I what representation is and how this thing functions in this model so I know often times people talk about like what's the explanatory value of talking in terms of representations um but I mean to answer more directly um what's so what's the alternative to it being structural that's another good question um in my view this structural representationalist Paradigm really isn't anything new it's kind of just like the core notion of representation that's been at work in serious cognitive science since like Turing so like if you

go back to um you know papers uh like um
there's a paper from 1980 by uh Alan new
on um sort of uh foundations of
cognitive science and and he he
described how um what makes a particular
symbolic um physical symbol system um
able
to uh what gives it its Universal
Universal computational capacity is its
capacity to simulate other systems so I
really think this notion of simulation
is at the heart of of um what
representation is supposed to be in
cognitive science and that that really
just is structural representation and if
you go back last I won't go on forever
if you go to um cummins's early work on
on structural representationalism um the
S representation can also be read as
simulation um so it's not that I think
that you need to call it a gen the
generative model a structural
representation I just think once you
understand how it functions um I would
say even under the FP that's like what
it is but there's still a question about
how it's encoded or whether it's encoded
I think that's a separate question a
awesome thanks for that response we'll
go to Lee and then anyone else that can
raise their hand okay so um sorry excuse
me I I was just trying to make sure that
um I I was understanding terms in the
same way that they were being used but
Alex you you just talked about the uh
the notion of representation simulation
and um I you know one of the ways I've
kind of arrived at active inference is

is via the work of of barel and
perceptual symbol Theory and simulation
does that is is that overlapping with
what you're meaning by simulation or
does it have a

different um meaning if you're familiar
with his work to probably I'm not I'm
not as familiar with that work um as
with um the the newal paper that I just
that I mentioned I am and it is just a
yeah

great no just kidding I want to read it
um okay I'll accept that I'll I'll take
next word for it yeah I mean essentially
uh especially these deep generative
models are they allow you to entertain
uh well I mean the there's a sense in
which the like the baralo you know idea
of metaphor is just what the generative
models do uh you know if we're talking
about a kind of loose associative
structure that redeploys inferences
about one domain in another I mean uh
this is the kind of thing that you would
expect a generative model would be able
to do uh so these things are definitely
I think strongly aligned uh in effect uh
I mean didn't step suggests something
similar I think uh like the yeah
metaphors are generative models in a
strong sense cool notion will go to Alex
kefir and then anyone
else yeah so um that's interesting I
don't know what to say about metaphors
but

um I think the way I see the way I've
this now is that um I don't think what
Maxville at all are saying is is at all

of course um maybe not of course I don't think it's at all wrong but I just think that our uh the sort of the structural representationalist reading my view is that it doesn't quite say enough but not that not that it's wrong in any particular respect as applied to the FP like so so after further conversations with Maxwell and also with um with Michael and the other authors on this um it seems to me that you still there's still a need for a sort of of neuronally realized generative model uh in addition to considering the entire phenotype to be a generative model so if you have a fancy organism that can do you know deep temporal kind of modeling and plan for the future and things like that um I think there's still a need for something that looks like a structural representation um in the brain that that is the generative model can be construed that way as well as there's this sort of larger more encompassing system and maybe this could be some somewhat hashed out by appealing to the multi-level active infant stuff um but last piece of this um in discussions with with um Michael kirov I hope I'm saying his name correctly we've never said it out loud to each other um it it seems as though he at least he is thinking of top- down sort of propagation of signals as a form of action so the stuff that we were saying about um top down generative models might just be sort of a special

case of the sort of action that um this paper is talking about cool before we go to Stephen and then Lee it really reminds me about Alejandra's point one or two weeks ago about what are the similarities and the differences between the systems that can at least appear to do these deep simulations counterfactuals like the brain versus a embodied phenotype the cell perimeter is not simulating other cell perimeters um that's a little bit different than the way that the brain might be able to do something strategic so um we'll go to Stephen and then anyone else can raise their hand yeah one one thing I always look at when you think of rep we talk about representation also in performance um is to think of like representation like is there another presentation in the brain which often people start thinking of as a projection and then you start getting into into this very visually dominated view of what our knowing is you know so whereas it's a bit different to reenacting or reimagining so like if I was to reimagine as an engineer um a a model and then I manipulate it in my brain well am I manipulating in my brain or have I reimagined it in my peripersonal space and am I now manipulating it at actively in my peripersonal space and that is what I'm actually thinking of as a representation but it's not actually my brain at all it's just a

reenactment in space for me to then
start to work with or if I do it on
paper I think the mechanism for that is
this is not quite clear but this sort of
opens up that question Cool Lee
uh Stephen I think I understand what
you're talking about but I are you just
kind of um mixing up the
phenomenological with the with the the
underlying ontological level so if you
have a representation in the brain and
you simulate it that it that it manifest
in experiential space or perceptual
space is that what you
mean well to to manifest it in the
phenomenological space is to enact is
what I'm saying so it's not like it's in
the brain then I project it into space
it's like it's not in the brain it might
be encoded like Place cells grid cells
all of that stuff with all the other
stuff going but it might be encoded in a
way that can reenact it but once it's
reenacted so with the work with clean
language and micro phenomenology looks
at like re embodying Concepts in space
around us so my kind of feeling is that
a metaphor is kind of like our reading
on our kind of affect and our kind
of thinking our beliefs recreated in a
kind of a physicalized way you know so
we actually do like we I feel like I'm
stuck behind a wall like I st you can
actually ask people well where's the
wall and they they'll know where the
wall is and what's it like to be stuck
and where so suddenly actually find that
they've actually got

like a model but they never had that model before they kind of created it so the question becomes is it a reenactment a reimagination and how much of that was already I suppose this is a question was that already sitting in the brain ready to go or did it only come about through the act of reimagining cool um in our last 10 minutes it would be awesome to hear from anyone who hasn't spoken as well as just to flip through the figures so that we can see where we're going to be headed to do some technical unpacking especially next week so anyone can feel free to raise their hand if they have a related or unrelated point but in these last few minutes let's just try to look at the figures and hear some closing thoughts so in figure one we have the marov blanket and active inference and we've returned to this figure many times it's used across papers and it's really nice to see it again and again because the context is always different and the part that I underlined in red was um in graph theoretic terms the marov blanket per se is defined as the set of nodes that isolates internal nodes from the influence of external ones and in in this capacity it's really good that we had the discussion about integrating internalism and externalism because those are also sort of like two ideas that have this tension between them and the Beyond internalism and externalism paper was about building on top of those

about taking the strengths of those two ideas and then asking how they're integrated under actm and today what we're doing and in this paper is about taking interpretations of internalism and externalism specifically this basian more computational perspective and this inactive embodied perspective and building on those strengths so it almost reads like a recipe for how active inference can be used for synthesis which is by identifying uh areas of tension in in the literature or previously unrecognized harmonies between different ideas and then just asking how we can build strength on strength rather than attack strength to weakness or just um you know contrast weaknesses and I'm just going to continue just to show the other figures but I'll pause anytime someone raises their hand on figure two oh Alejandra go ahead uh actually yeah in that figure um I can take again uh the question you mentioned um I'm still confused in terms of um this generative models and their recognition density um in which temporal and um spal scale um OCC course in each individual cell of the brain this this process is occurring and then in I don't know in a layer of the cortex and then in the the hierarchy um of of the cortical layers and then in the whole brain and then blah blah blah but um so um belief updating is happening in each cell um I don't know if if um

if if I'm
um I'm I'm getting to to to to the to
the point but um the cells have to
maintain actually this blanket contains
also some beliefs that never can change
right um those so in to to continue
being the the cell it it is um um so um
I don't know uh
this this this process of of the
densities is is
occurring in I don't know how many
scales and that's all of that I mean uh
we'll we'll end up uh hopefully
discussing this next week if it's if
it's unclear I mean I have to say I
don't think the figures are super useful
in this paper uh they're mostly there to
illustrate a few things what's really
important is the equations
um so I mean for next week Dan I'll show
you which ones in particular so that we
can put them up uh yeah so uh Alejandra
what you were saying is exactly right
all of these things are performing a
belief updating and when we talk about
Behavior or Dynamics or whatever it it
is all belief updating and belief
updating just means changing the
physical value of the recognition
density at a moment so basically um in
this paper like the big kind of
conceptual contribution that I tried to
make to this literature is to say well
uh with active inference we can rethink
what we mean by embodiment and enactment
so embodiment means encoding a
recognition
density uh so your physical body is a

guess about the it's a posterior probability uh that you're you're constantly updating through this process of belief updating AKA Dynamics AKA adaptive Behavior like you've got this we live literally embody this this density uh the physical states of our body encode the parameters of probability densities this is like the this is the idea um and then basically the the generative model is just the point of reference for the Dynamics that exists in the Dynamics the way I like to think about it is sort of like dominoes falling over where the recognition density is The Dominoes right each of these dominoes is like a posterior it you know and they kind of accumulate but the wave is the generative model you see like the Domino wave itself that that the generative model exists in the same sense as uh the wave kind of you know making all the The Dominoes fall and the recognition model exists in the sense of the the physical dominoes that are kind of encoding the the process as it kind of flows over uh but this will be I think hopefully clearer next week mathematically it's it's it's actually pretty simple it's that the generative model is the joint probability density over all of your variables and this this density doesn't exist in the brain it's not anywhere uh what exists is the the factorization of this density so you know the you always see this in all these active inference papers it's like P of all of your

variables so Ada mu and all the parameters Etc equals and then this long factorized which is basically a it's it's a product of uh likelihoods and priors so this product of likelihood and priors is the recognition density that's constantly updated right and that's what exists in the brain uh but this joint density only exists as a function of the Dynamics it's like all of these part partial uh like carving UPS of this density right this factorization they all move together and in their Dynamics together they realize this joint density sorry I just I I do have to end it at the 59 as we discuss but this is the perfect excitement to build for our follow-up discussion next week we're definitely going to go into this question we'll just pick up right here we'll literally just hit play on the video and uh go into detail on the technicality and let's even have some dominoes on the screen so again just thanks everybody for understanding about the timing I've provided a follow-up form to the live participants in the chat and we welcome any other feedback suggestions or questions and please just uh stay in touch stay engaged if this was interesting or exciting we welcome all participants and it would be awesome to have you on this stream asking the questions and also learning by doing with us so thanks everyone for the awesome and energizing discussion and

I'm really looking forward to next week
when we can go another level detail into
all of this so thanks so much